

VIRTUALIZATION

An Introduction

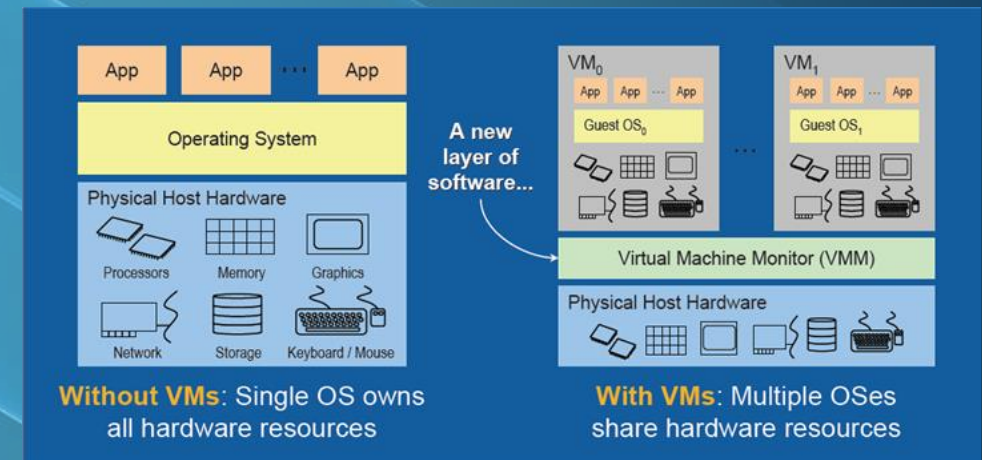


- ▶ Definition
- ▶ Types
- ▶ 5 Key Components
- ▶ Benefits / Limitations
- ▶ Cloud vs Virtualization
- ▶ Cloud Service Models

AGENDA



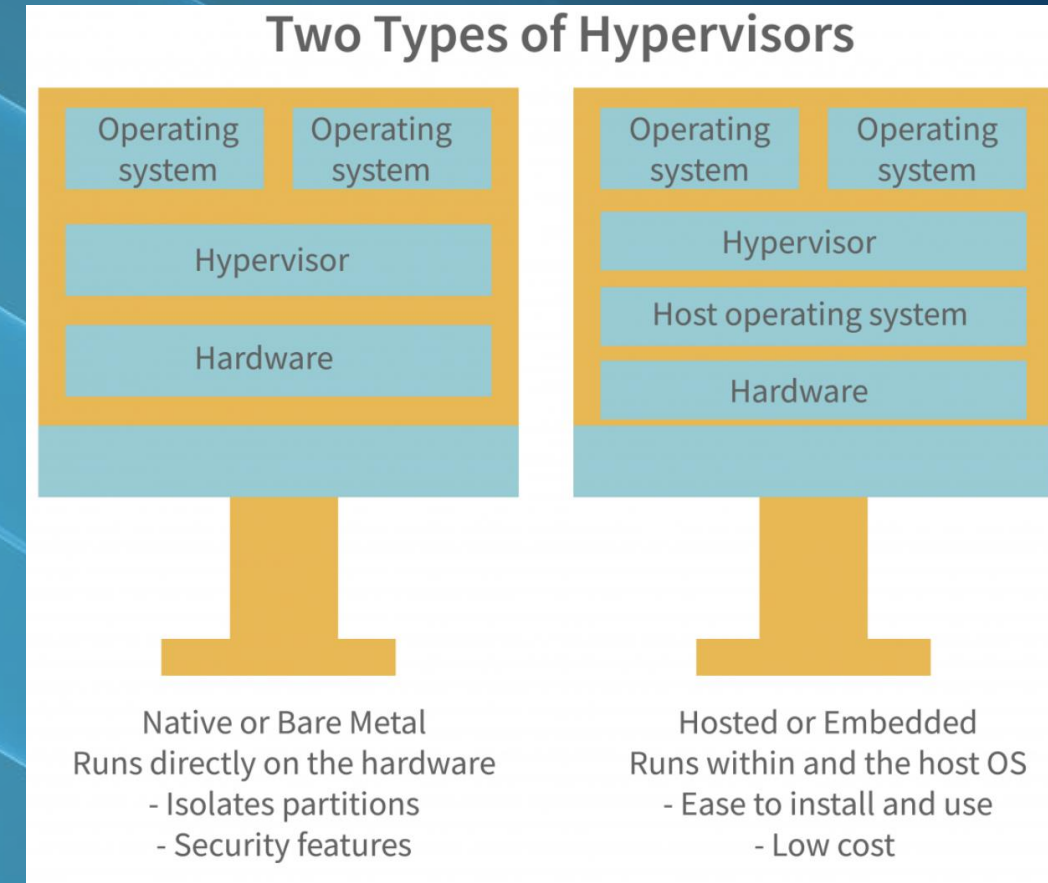
- ❑ The provision of a set of computing resources or their logical combination, abstracted from the hardware while providing logical isolation of computing processes running on the same physical resource
- ❑ Virtualization is realized with the help of software – a hypervisor
 - ❑ **A virtualization host** is a physical server with a hypervisor running on it.
 - ❑ **A host system** is the physical machine OS that has the installed hypervisor.
 - ❑ **A guest operating system** is an OS running inside the virtual machine.
 - ❑ **A virtual machine** is a virtual copy of real hardware, an application that functions as a physical computer or server.



DEFINITION



- ❑ Full Virtualization **aka** Native / Bare Metal
 - ❑ Interacts directly with the physical server
 - ❑ Serves as a platform for the virtual server's operating system
 - ❑ Considered as Type 1 Hypervisor
- ❑ Para Virtualization **aka** Hosted / Embedded
 - ❑ Interacts with Hypervisor
 - ❑ Host Operating is a must
 - ❑ Considered as Type 2 Hypervisor



VIRTUALIZATION TYPES



- ❑ Server Virtualization
- ❑ Desktop Virtualization
- ❑ Application Virtualization
- ❑ Storage Virtualization
- ❑ Network virtualization

5 KEY COMPONENTS



❑ Benefits

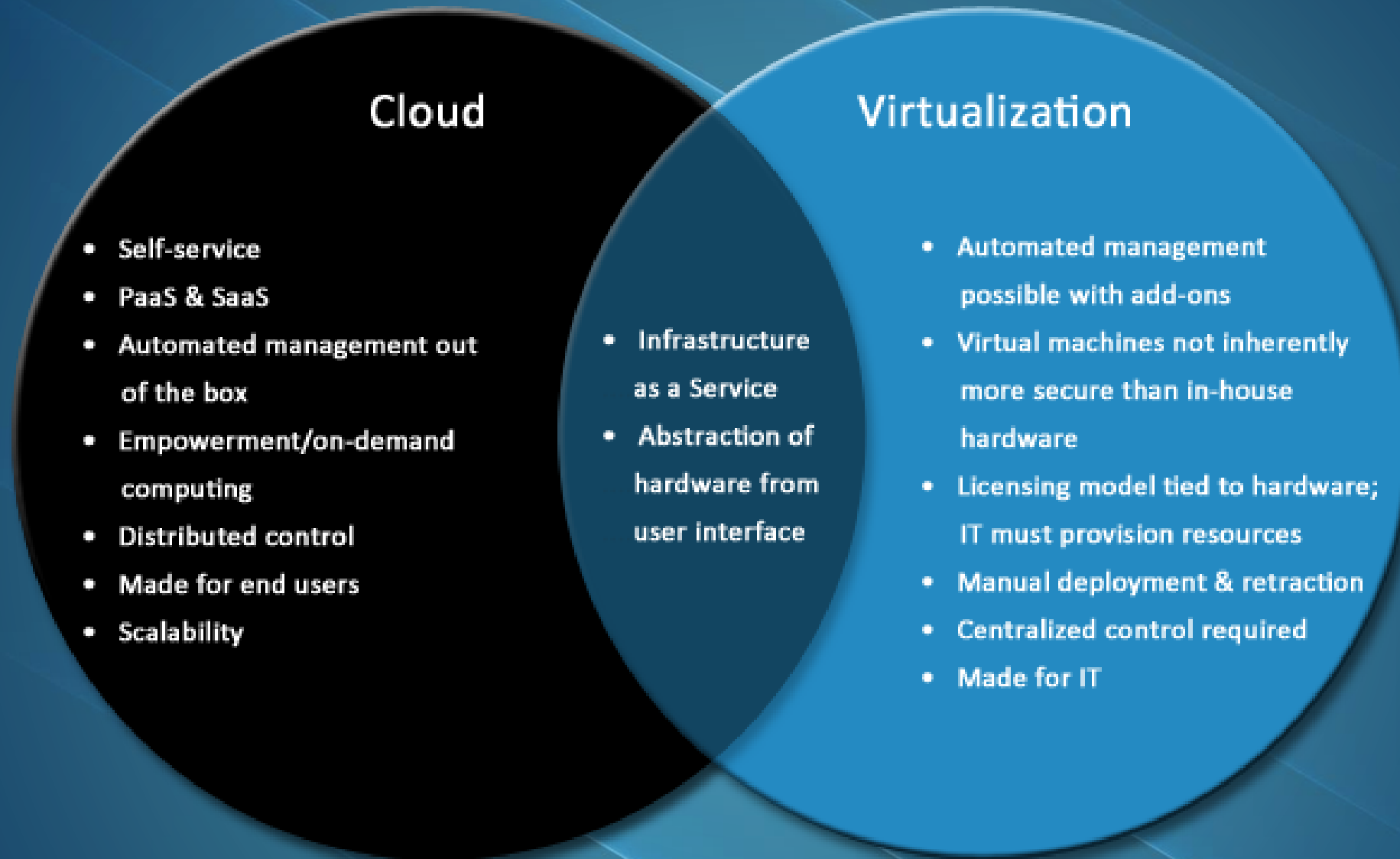
- ❑ Server Consolidation
- ❑ Data Replication
- ❑ Resource Optimization
- ❑ High Fault Tolerance
- ❑ Flexibility
- ❑ Cost Effectiveness

❑ Limitations

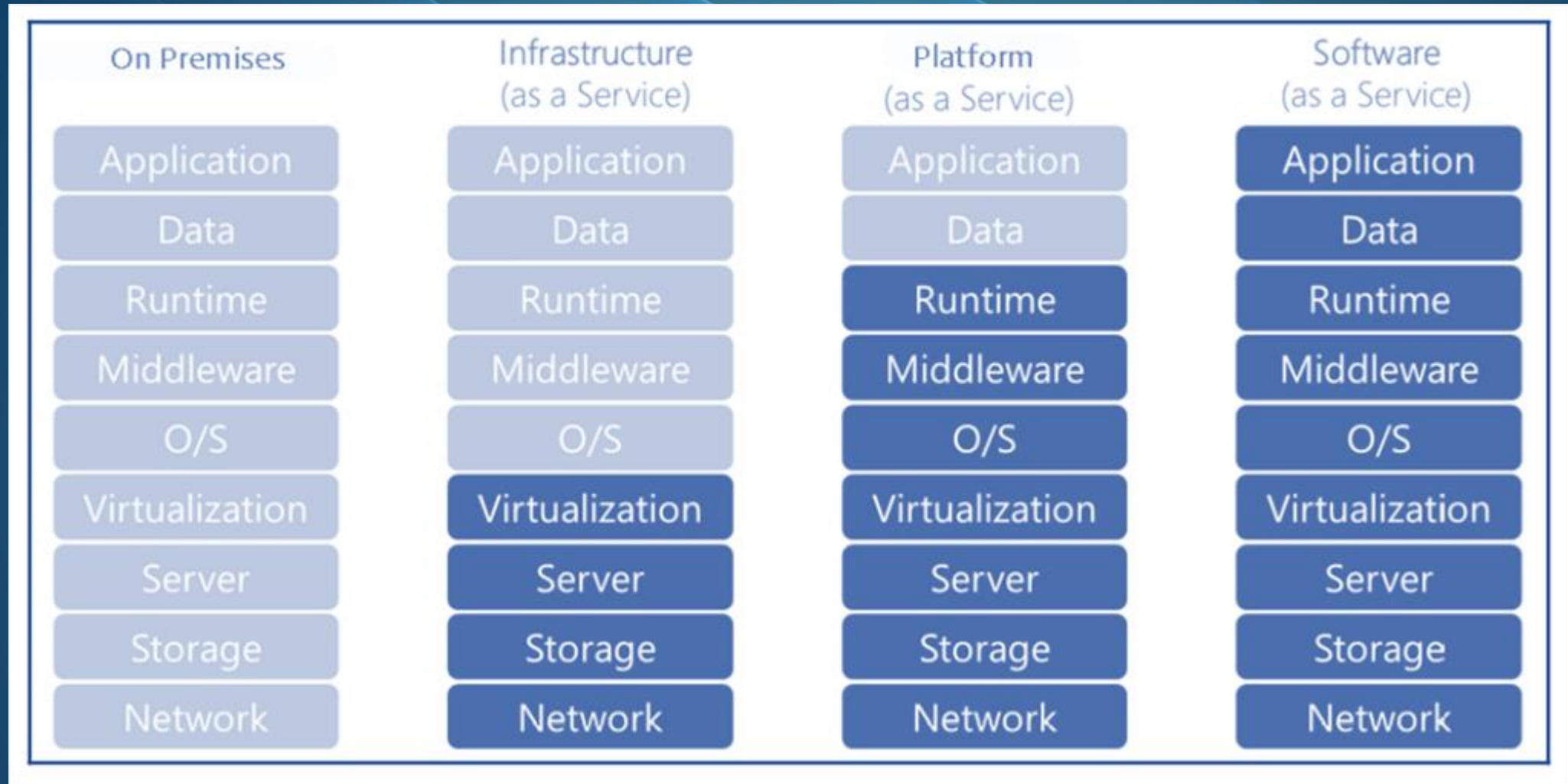
- ❑ Performance Overhead
- ❑ Resource Contention

BENEFITS / LIMITATIONS

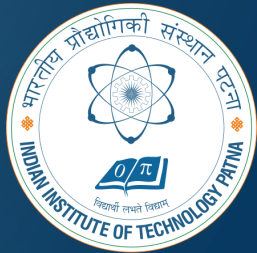


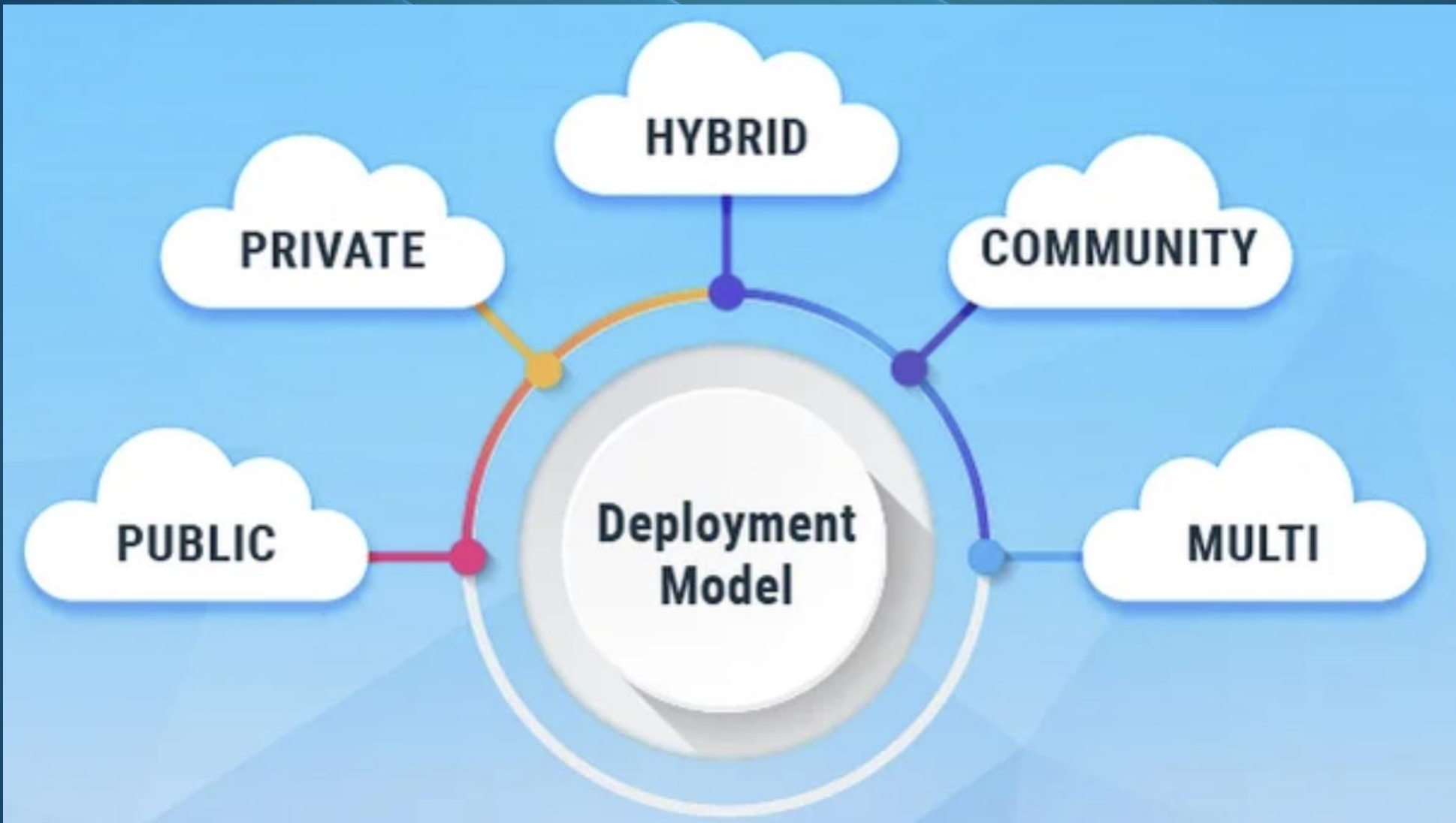


CLOUD VS VIRTUALIZATION

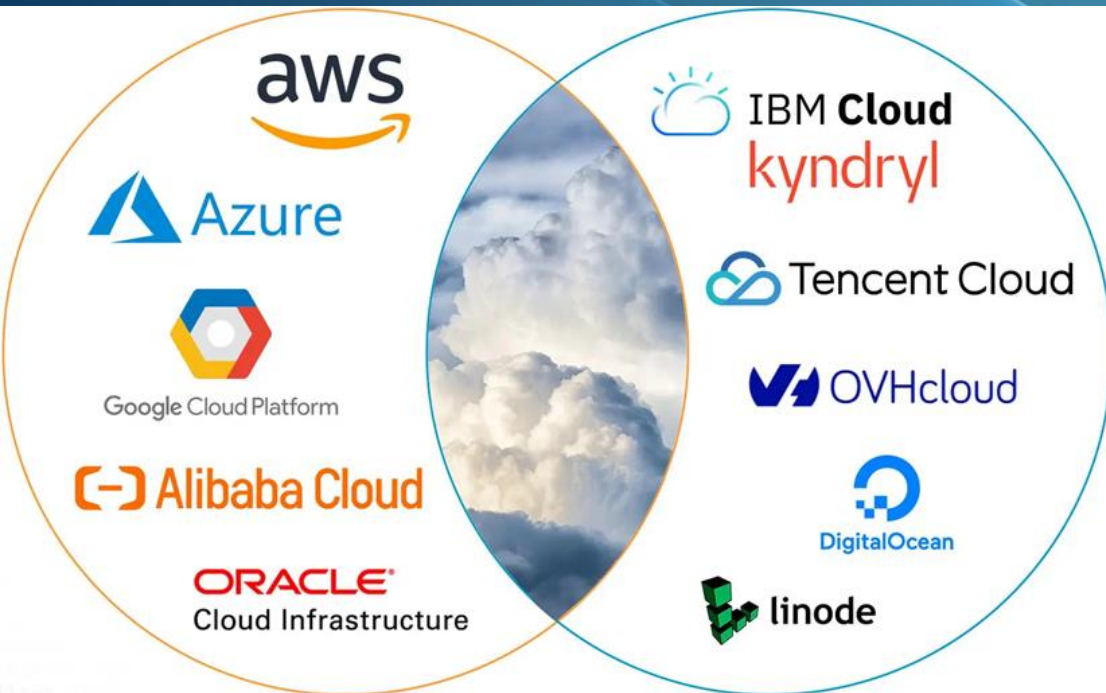


CLOUD SERVICE MODELS





CLOUD DEPLOYMENT MODELS



#	Cloud Service Provider	Regions	Availability Zones
1	Amazon Web Services (AWS)	33	105
2	Microsoft Azure	64	126
3	Google Cloud Platform (GCP)	40	121
4	Alibaba Cloud	30	89
5	Oracle Cloud	48	58
6	IBM Cloud	10	30
7	Tencent Cloud	21	65
8	OVHcloud	17	37
9	DigitalOcean	9	15
10	Linode (Akamai)	20	20

TOP 10 CLOUD SERVICE PROVIDERS

- ❑ A computer system created using software on one physical computer in order to emulate the functionality of another separate physical computer
- ❑ A compute resource that uses software instead of a physical computer to run programs and deploy apps
- ❑ One or more virtual “guest” machines run on a physical “host” machine
- ❑ Each virtual machine runs its own operating system
- ❑ Functions separately from the other VMs, though they are all running on the same host
- ❑ Used for many use cases across on-premises and cloud environments
- ❑ Public cloud services are using virtual machines to provide virtual application resources to multiple users at once, for more cost efficient & flexible compute



VIRTUAL MACHINE - DEFINITION



- ❑ Using Secure Shell
- ❑ Using Systems Manager Session Manager
- ❑ Using Remote Desktop Protocol (RDP)
- ❑ Accessing VMs via Elastic IPs and Public DNS
- ❑ Using Bastion Hosts
- ❑ Accessing VMs from CLI
- ❑ Accessing VMs from Cloud Shell
- ❑ How-to
 - ❑ Linux Instances
 - ❑ Use SSH for direct access, or Systems Manager Session Manager for access without SSH keys.
 - ❑ Windows Instances
 - ❑ Use RDP for remote desktop access, or Systems Manager Session Manager for access without RDP.
 - ❑ Elastic IPs
 - ❑ Provide a static IP for consistent access.
 - ❑ Bastion Hosts
 - ❑ Facilitate access to private instances within a VPC.
 - ❑ CLI/Cloud Shell:
 - ❑ Manage instances and perform actions via command-line tools.

ACCESSING VIRTUAL MACHINES



DEMONSTRATION

Launching and accessing a virtual machine in AWS

HANDS-ON EXERCISE

- Create a Free Tier account in Azure, if not already
- Launch a virtual machine in Azure using UBUNTU
- Configure a SAMPLE web site
- Access the website (Internal & External)



THANK YOU !

